



Access S3 from a VPC



yakubu abdullahi yusuf

```
[ec2-user@ip-10-0-9-194 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-9-194 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-sthree
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-sthree/test.txt
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls
2026-01-15 22:49:05 nextwork-vpc-project-sthree
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls s3://nextwork-vpc-project-sthree
2026-01-15 22:51:14      2431554 NextWork - Denzel is awesome.png
2026-01-15 22:51:15      2399812 NextWork - Lelo is awesome.png
2026-01-15 23:11:24           0 test.txt
[ec2-user@ip-10-0-9-194 ~]$
```



yakubu abdullahi yusuf

NextWork Student

nextwork.org

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service that lets you create your own private network inside AWS where you can run cloud resources like EC2 instances. It is useful because it gives you full control over networking and security.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a VPC, launch an EC2 instance in the VPC and connect to an S3 bucket through EC2 connect and also uploaded a file through AWS CLI to the S3 bucket.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how fun it is to use AWS CLI to access different AWS services.

This project took me...

This project took me 30 minutes to complete.



yakubu abdullahi yusuf

NextWork Student

nextwork.org

In the first part of my project...

Step 1 - Architecture set up

In this step, I will create a VPC and Launch an EC2 instance into the VPC.

Step 2 - Connect to my EC2 instance

In this step, I will be connecting to my EC2 instance.

Step 3 - Set up access keys

In this step, I will be creating access keys to allow access to services through AWS CLI.



yakubu abdullahi yusuf

NextWork Student

nextwork.org

Architecture set up

I started my project by launching a VPC with one subnet and then Launched an instance in the VPC.

I also set up an S3 bucket and uploaded some files in it.

Summary

Destination

s3://nextwork-vpc-project-sthree

Succeeded

2 files, 4.6 MB (100.00%)

Failed

0 files, 0 B (0%)

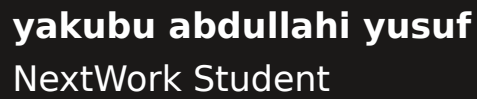
Files and folders

Configuration

Files and folders (2 total, 4.6 MB)

Find by name

Name	Folder	Type	Size	Status	Error
NextWork - Denzel is awesome...	-	image/png	2.3 MB	Succeeded	-
NextWork - Lelo is awesome.pn...	-	image/png	2.3 MB	Succeeded	-



The first command I ran was "aws s3 ls" This command is used to list all the S3 an account.

The second command I ran was "aws configure" This command is used to set up credentials so the CLI can interact with an AWS account.

```

#
#####
#####|
#####|
#/\
V- -> https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls

[ec2-user@ip-10-0-9-194 ~]$ aws configure
AWS Access Key ID [None]: 

```



yakubu abdullahi yusuf

NextWork Student

nextwork.org

Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured my AWS CLI with AWS Access Key ID and AWS Secret Access Key, region name and output format.

Access keys are a set of credentials that allow access to AWS services via the CLI,.

secret access key is like the password that pairs with your access key ID to access AWS services.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use AWS CloudShell, a browser-based CLI, to run commands. Use the AWS CLI V2 to enable authentication through a user in IAM identity center.



yakubu abdullahi yusuf

NextWork Student

nextwork.org

In the second part of my project...

Step 4 - Set up an S3 bucket

In this step, I will create an S3 bucket.

Step 5 - Connecting to my S3 bucket

In this step, I will be interacting with my S3 bucket through my EC2 instance.



Connecting to my S3 bucket

The first command I ran was "aws s3 ls" This command is used to list all the S3 an account.

When I ran the command aws s3 ls again, the terminal responded with the name of my s3 bucket. This indicated the access key configuration worked and s3 is available, and I can access AWS services through AWS CLI.

```
default output format [None]: [ec2-user@ip-10-0-9-194 ~]$ aws s3 ls
026-01-15 22:49:05 nextwork-vpc-project-sthree
ec2-user@ip-10-0-9-194 ~]$
```



Connecting to my S3 bucket

Another CLI command I ran was `aws s3 ls s3://nextwork-vpc-project-sthree` which returned the objects in my S3 bucket.

```
Default output format [None]: [ec2-user@ip-10-0-9-194 ~]$ aws s3 ls
2026-01-15 22:49:05 nextwork-vpc-project-sthree
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls s3://nextwork-vpc-project-sthree
2026-01-15 22:51:14      2431554 NextWork - Denzel is awesome.png
2026-01-15 22:51:15      2399812 NextWork - Lelo is awesome.png
[ec2-user@ip-10-0-9-194 ~]$
```



Uploading objects to S3

To upload a new file to my bucket, I first ran the command `sudo touch /tmp/test.txt`. This command creates a new file.

The second command I ran was `aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-sthree`. This command will copy the file `test.txt` into my S3 bucket.

The third command I ran was `aws s3 ls s3://nextwork-vpc-project-sthree`, which validated that the file was successfully copied to the S3 bucket.

```
[ec2-user@ip-10-0-9-194 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-9-194 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-sthree
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-sthree/test.txt
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls
2026-01-15 22:49:05 nextwork-vpc-project-sthree
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls s3://nextwork-vpc-project-sthree
2026-01-15 22:51:14      2431554 NextWork - Denzel is awesome.png
2026-01-15 22:51:15      2399812 NextWork - Lelo is awesome.png
2026-01-15 23:11:24           0 test.txt
[ec2-user@ip-10-0-9-194 ~]$
```



nextwork.org

The place to learn & showcase your skills

Check out nextwork.org for more projects

